Deggendorf Institute of Technology
Campus Cham

Course Name

MASTER THESIS

Sensor Fusion of Camera and Pressure Sensor Data for Cabin Monitoring System

submitted in the requirements for the degree of

Master of Engineering

by

CHETHAN SRINIVASAREDDY

Matriculation Number:

Supervisor: [SUPERVISOR NAME]

Cham, 2026

Acknowledgements

I would like to express my sincere gratitude to my supervisor, [SUPERVISOR NAME], for the continuous guidance, constructive feedback, and support provided throughout the course of this thesis. I would further like to thank the research group at Deggendorf Institute of Technology, Campus Cham, for providing access to the laboratory infrastructure used for pressure sensor and camera data collection. Finally, I am grateful to my family and friends for their patience and encouragement during this work.

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[CHETHAN SRINIVASAREDDY]

Abstract

In-cabin occupant monitoring is increasingly required by vehicle safety regulation and by advanced restraint systems that must adapt to who is seated where. This thesis investigates the fusion of RGB camera images with pressure heatmaps obtained from seat-embedded pressure sensors in order to jointly estimate seat occupancy, passenger class, occupant weight, and body pose. A synthetic pressure heatmap generation pipeline based on human body simulation is proposed to complement limited real sensor data, together with a Sim2Real validation procedure that quantifies the remaining domain gap. A multi-task deep learning fusion architecture combining camera and pressure branches is designed, trained, and evaluated against single-modality baselines through ablation studies. The expected contribution is a validated fusion pipeline and dataset methodology that improves robustness of occupant state estimation over either sensing modality used in isolation.

Keywords: cabin monitoring, sensor fusion, pressure heatmap, occupant classification, pose estimation, Sim2Real, multi-task learning

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List of Abbreviations

Abbreviation	Meaning
CNN	Convolutional Neural Network
DMS	Driver Monitoring System
EKF	Extended Kalman Filter
FOV	Field of View
GDPR	General Data Protection Regulation
IMU	Inertial Measurement Unit
IR	Infrared
MSE	Mean Squared Error
OMS	Occupant Monitoring System
RGB-D	Red-Green-Blue plus Depth (image with depth channel)
ROI	Region of Interest
Sim2Real	Simulation-to-Reality (transfer)
ToF	Time-of-Flight
UKF	Unscented Kalman Filter
WRMSE	Weighted Root Mean Squared Error

1 Introduction

1.1 Motivation and Relevance

Occupant monitoring is becoming an increasingly important component of modern vehicle safety systems. Traditionally, vehicle restraint systems were designed around relatively constrained assumptions: occupants were expected to be seated in approximately upright, forward-facing positions, and the driver was expected to remain continuously engaged in the driving task. These assumptions are becoming less representative as vehicles incorporate higher levels of driving automation, more configurable interiors, and a wider range of in-cabin activities. In highly automated vehicles, occupants may increasingly recline, rotate their seats, interact with other passengers, use mobile devices, work, or perform other non-driving-related activities. Consequently, the vehicle must account not only for whether a seat is occupied, but also *who or what occupies the seat and how the occupant is positioned* [**mcmurry2018oop**, **reichert2022rotated**].

This change in occupant behaviour is particularly relevant to crash protection. Existing restraint systems, including three-point seat belts and frontal airbags, have predominantly been developed and validated for nominal seating positions. Out-of-position occupants can interact with these restraint systems in substantially different ways. McMurry et al. showed that reclined and turned or sideways postures are already observed in real-world crashes and highlighted the potential safety challenges associated with such postures in highly automated vehicles [**mcmurry2018oop**]. More importantly, analysis of real-world crash data from the German In-Depth Accident Study (GIDAS) found significantly increased injury risk for reclined occupants. Schaefer et al. reported odds ratios of approximately 2.07, 3.09, and 3.66 for MAIS 2+, MAIS 3+, and MAIS 4+ injuries, respectively, for belted reclined occupants compared with belted occupants in upright seating positions [**schaefer2021reclined**]. Simulation studies considering reclined and rotated seats similarly demonstrate that occupant orientation and seatback angle can substantially alter occupant kinematics, restraint interaction, and predicted injury risk [**reichert2022rotated**]. These findings indicate that future restraint strategies cannot necessarily assume a fixed occupant posture.

The requirement to understand occupant state also extends to conditionally automated driving. At SAE Level 3, the driver may temporarily disengage from the dynamic driving task but must remain capable of responding to a transition demand. Experimental investigations have demonstrated that non-driving-related postures, particularly large trunk deviations, unconventional leg positions, or activities occupying both hands, can increase takeover time and reduce takeover quality [**zhao2024takeover**]. Therefore, information about driver posture, presence, and readiness becomes relevant not only to passive crash protection but also to the safe operation of automated-driving functions.

Airbag deployment provides a particularly strong example of why accurate occupant information is required. Frontal airbags are highly effective safety devices; the National Highway Traffic Safety Administration (NHTSA) estimates that they have saved more than 50,000 lives over approximately three decades [**nhtsaAirbags**]. Nevertheless, an airbag is also a rapidly deploying restraint device and can itself cause severe or fatal injuries when an occupant is very close to the deployment module, is small in stature, is incorrectly restrained, or is otherwise out of position. Historical crash investigations illustrate the significance of this problem. By January 2003, NHTSA's Special Crash Investigation programme had identified 227 confirmed fatalities in which an airbag deployment caused a fatal injury during a minor-to-moderate crash; 141 of these confirmed cases involved children [**nhtsaAirbagFatalities**]. Earlier CDC/NHTSA investigations reported 32 child fatalities associated with passenger-airbag deployment between January 1993 and November 1996, including infants placed in rear-facing child restraint systems in front of an active passenger airbag [**cdc1997airbags**]. Although these statistics primarily concern earlier generations of airbag technology, they demonstrate the fundamental safety problem that motivated the development of occupant-sensitive restraint systems.

This problem contributed directly to the development of advanced airbags and occupant classification systems. In the United States, the May 2000 amendment to Federal Motor Vehicle Safety Standard (FMVSS) No. 208 introduced advanced-airbag requirements intended to reduce airbag-induced injuries to children and small adults while preserving frontal-crash protection for other occupants [1]. Modern advanced restraint systems may use information from crash-severity sensors, seat-belt sensors, seat-position sensors, and occupant-weight or classification sensors to determine whether an airbag should deploy and, in systems equipped with multiple inflation stages, the appropriate deployment level [**nhtsaAdvancedAirbag**]. Thus, occupant sensing is not merely a comfort-related function: inaccurate classification may influence safety-critical decisions such as passenger-airbag suppression or the selection of an appropriate restraint deployment strategy.

European regulations provide the broader crashworthiness context. UN Regulation No. 94 addresses occupant protection in frontal collisions, while UN Regulation No. 95 addresses protection in lateral collisions [**uneceR94**, **uneceR95**]. These regulations should not be interpreted as directly mandating an occupant monitoring system; rather, they establish the crash-protection environment within which restraint systems must perform. More recent regulations associated with automated driving establish a more direct connection to cabin monitoring. UN Regulation No. 157 concerning Automated Lane Keeping Systems requires a Driver Availability Recognition System capable of assessing whether the driver is present and available to resume the driving task when necessary [**uneceR157**]. Similarly, the European Union's General Safety Regulation and Commission Delegated Regulation (EU) 2023/2590 establish requirements for advanced driver-distraction warning systems [**eu2023addw**]. These developments demonstrate a regulatory transition from systems concerned mainly with vehicle dynamics and crash response toward systems that also continuously assess the human state inside the vehicle.

Consumer safety assessment is moving in the same direction. From 2026, Euro NCAP places increased emphasis on occupant and driver monitoring as part of its Safe Driving assessment. Its framework rewards systems capable of continuous driver monitoring, correct seat-belt-use detection, and adaptation of restraint and airbag functions for different occupant physiques. The 2026 framework additionally requires reliable determination of the number of vehicle occupants for automated post-crash emergency information [**euroncap2026**]. Consequently, reliable estimation of occupancy, occupant characteristics, and posture is becoming relevant across multiple stages of vehicle safety: before a crash, during restraint deployment, and following a collision.

Achieving such reliable occupant-state estimation using a single sensing modality remains challenging. Camera-based systems provide rich spatial and semantic information and can estimate body posture, keypoints, head orientation, and visually distinguish between people and objects. However, vision-based monitoring is affected by variations in illumination, glare, shadows, camera viewpoint, partial body visibility, and occlusion [**salazar2025dms**]. Occlusion is particularly relevant inside a vehicle because parts of an occupant may be hidden by seats, other passengers, clothing, blankets, carried objects, or the vehicle structure. These conditions can reduce the reliability of image-based pose and occupant classification even when the camera itself remains operational.

Seat-embedded pressure sensing provides a fundamentally different source of information. Pressure distributions directly represent the mechanical interaction between an occupant and the seat and are therefore independent of visible-light conditions. Previous work has demonstrated that seat-pressure distributions contain information useful for occupant classification and body-weight estimation [**reed2001ocatd**]. However, pressure measurements are not semantically descriptive in the same manner as images. Similar total loads can be produced by different occupants or objects, and pressure distribution varies with body morphology, seating posture, seat geometry, and the placement of the load. NHTSA specifically notes that advanced passenger-classification systems based on sensed seat loading can produce unexpected classifications when external conditions add or remove load from the seat [**nhtsaAirbags**]. Pressure sensing alone may therefore provide strong evidence of physical occupancy, contact distribution, and load while remaining ambiguous regarding the visual identity of the occupant or object and some aspects of body configuration.

Camera and pressure sensing are consequently complementary rather than redundant modalities. The camera provides information about *visual appearance and geometric configuration*, whereas the pressure mat provides information about *physical contact and loading*. A condition that degrades one modality does not necessarily degrade the other: poor illumination does not alter the physical pressure distribution on the seat, while ambiguous pressure distributions may be resolved using visual evidence. Multimodal learning can exploit such complementarity, correlation, and partial redundancy to improve robustness compared with models relying on a single sensor stream [**ortega2022multimodal**].

This complementarity forms the central motivation of the present thesis. The work investigates whether jointly learning from camera images and seat-pressure maps can provide more reliable estimates of occupant state than independently trained camera-only or pressure-only systems. The occupant state considered in this work includes seat occupancy, occupant class, posture, and occupant weight or weight category. By systematically comparing the individual modalities with their fused representation, the study aims to quantify the actual benefit of heterogeneous sensor fusion and determine which occupant-monitoring tasks benefit most from the complementary information provided by vision and pressure sensing. The broader objective is to contribute toward cabin-monitoring systems that remain reliable across variations in occupant morphology, seating posture, activity, illumination, and partial sensor degradation, thereby supporting future adaptive restraint and automated-vehicle safety functions.

1.2 Problem Statement

Existing occupant classification systems typically rely on a single sensing modality, which limits their robustness across the wide range of seating configurations, occupant sizes, and environmental conditions encountered in real vehicles. Furthermore, collecting sufficiently large and diverse real-world pressure sensor datasets is costly and raises data protection concerns, motivating the use of synthetic data generation. This thesis addresses the problem of designing, training, and evaluating a multi-task fusion model that jointly estimates occupancy, passenger class, weight, and pose from synchronised camera and pressure heatmap inputs. [2] [3] [4] [5]

1.3 Research Questions

The following research questions guide the design, implementation, and evaluation carried out in this thesis: " evaluvating suitability of heterogeneous fusion framework for the proposed modalities,comapring its peformance to single-modality baselines".

1. Fusion Benefits: To what extent does fusing camera and pressure heatmap data improve occupancy, passenger classification, weight estimation, and pose estimation accuracy compared to single-modality baselines?
2. Suitability of fusion architecture for these modalities?
3. How should synthetic pressure heatmaps be generated and validated to ensure they are statistically representative of real seat pressure sensor recordings (Sim2Real gap)? [6]
4. Which feature-level fusion strategy provides the best trade-off between accuracy and computational cost for a real-time cabin monitoring application?
5. degradation and abstention : how well can it peform under occlusion, lighting changes, faulty pressure sensor data and other adverse conditions, and can it abstain from making predictions when one of the modalities is uncertain/low quality input?

[7] this is a novel deep fusion architecture that can be used to answer the above research questions.

1.4 Objectives and Scope

The primary objective of this thesis is to design and evaluate a multi-task deep learning architecture that fuses camera images and pressure heatmaps for cabin occupant state estimation. The scope covers system requirements definition, data collection , synthetic pressure heatmaps generation generation via human body simulation, a preprocessing and synchronisation pipeline, the fusion architecture itself, and a quantitative evaluation including ablation studies.

1.5 Thesis Structure

Chapter 2 introduces the background and fundamentals of cabin monitoring systems, pressure sensing technology, camera systems, and sensor fusion theory required to follow the remainder of the thesis. Chapter 3 reviews related work on pressure-based and camera-based occupant monitoring, multi-modal fusion approaches, and synthetic data generation, concluding with a gap analysis that positions this work. Chapter 4 defines the use cases, functional and non-functional requirements, and overall system architecture. Chapter 5 describes the real data collection protocol, the synthetic pressure heatmap generation pipeline, and the Sim2Real validation procedure. Chapter 6 presents the preprocessing and temporal synchronisation pipeline that prepares camera and pressure data for model training. Chapter 7 details the multi-task fusion architecture, loss formulation, and training configuration. Chapter 8 reports quantitative results, ablation studies, and Sim2Real transfer evaluation. Finally, Chapter 9 summarises the contributions, answers the research questions, discusses limitations, and outlines future work.

2 Background and Fundamentals

2.1 Cabin Monitoring Systems

Cabin monitoring systems are commonly divided into driver monitoring systems (DMS), which focus on the driver's attentiveness and drowsiness, and occupant monitoring systems (OMS), which assess the presence, position, and classification of all vehicle occupants. Regulatory bodies such as Euro NCAP have progressively increased the weight given to occupant status monitoring in their safety rating protocols, driving adoption of both camera-based and pressure-based sensing across vehicle segments [8]. Typical OMS use cases include seatbelt reminder logic, smart airbag suppression for child seats, and post-crash occupant status reporting for emergency services.

2.2 Pressure Sensing Technology

2.2.1 Working Principle and Sensor Types

Seat-integrated pressure sensors typically use resistive, capacitive, or piezoelectric transduction principles arranged in a grid to produce a spatially resolved pressure map of the seat surface. Resistive force sensing resistor (FSR) arrays are low-cost and widely used in automotive seat occupancy classification, while capacitive mats offer higher sensitivity at increased manufacturing cost [9]. The choice of sensor technology directly affects spatial resolution, sampling rate, and susceptibility to drift over the vehicle's operating temperature range.

[10] This is paper on lower resolution pressure sensor array for driver posture monitoring and pose estimation.

2.2.2 Pressure Heatmap Representation

The raw output of a pressure sensor grid is commonly represented as a two-dimensional heatmap, where each pixel encodes the normalised pressure value at a corresponding physical location on the seat surface. This representation allows pressure data to be processed with the same convolutional architectures typically applied to single-channel images, which is a key enabler for the fusion architecture proposed in Chapter 7.

2.3 Camera Systems for In-Cabin Use

2.3.1 RGB and IR Cameras

Standard RGB cameras provide high-resolution colour imagery suitable for detailed pose and appearance analysis but perform poorly under low-light or backlit conditions common in vehicle cabins. Near-infrared (IR) cameras paired with active IR illumination maintain consistent image quality irrespective of ambient lighting and are therefore preferred in production driver and occupant monitoring systems [11].

2.3.2 Depth Cameras

2.4 Sensor Fusion Fundamentals

2.4.1 Fusion Levels

Sensor fusion approaches are typically categorised into data-level fusion (combining raw signals before any feature extraction), feature-level fusion (combining learned intermediate representations), and decision-level fusion (combining independent per-modality predictions). Feature-level fusion generally offers the best balance between representational flexibility and robustness to modality-specific noise, and is the strategy adopted for the architecture in this thesis [12].

2.4.2 Deep Learning-Based Fusion

Deep learning-based fusion replaces hand-crafted process and measurement models with learned convolutional or attention-based feature extractors per modality, followed by a learned fusion module, building on the general deep learning foundations of representation learning and gradient-based training described in [13]. This approach scales naturally to high-dimensional, spatially structured inputs such as camera images and pressure heatmaps, and can be trained end-to-end jointly with the downstream task heads [14].

2.5 Multi-Task Learning

Multi-task learning trains a single shared backbone with multiple task-specific output heads, allowing related tasks to regularise one another and share statistical strength, which is particularly valuable when labelled data for some tasks (e.g. weight estimation) is scarcer than for others (e.g. binary occupancy). A central challenge in multi-task learning is balancing the loss contributions of tasks with different scales and convergence rates, which is addressed explicitly for this thesis in Section 7.6.

2.6 Human Body Modelling for Simulation

Parametric human body models represent body shape and pose using a low-dimensional set of shape and joint-angle parameters, enabling generation of anatomically plausible body meshes across a wide population of statures and weights. Combined with a physics-based seat contact simulation, such models allow synthetic pressure heatmaps to be rendered for arbitrary occupant configurations without requiring a physical subject, which forms the basis of the synthetic data generation pipeline described in Section 5.4 [15].

3 Related Work and State of the Art

3.1 Pressure-Based Occupant Detection and Classification

Early automotive seat occupancy detection relied on simple threshold-based binary weight switches, which were later extended to spatially resolved pressure mats capable of distinguishing occupant classes beyond a simple occupied/unoccupied signal [9]. Subsequent work applied regression neural networks directly to pressure heatmaps to estimate continuous occupant weight, demonstrating that spatial pressure distribution carries substantially more information than aggregate load alone [16]. A recent survey consolidates these pressure-sensing approaches and highlights classification accuracy, sensor cost, and robustness to seat cover variation as the primary axes along which such systems are compared [17].

3.2 Camera-Based Cabin Monitoring

Camera-based approaches to cabin monitoring have progressed from frontal-face driver drowsiness detection toward full-cabin, multi-occupant pose estimation. Infrared camera systems combined with real-time pose estimation pipelines have been shown to operate reliably across day and night conditions inside the vehicle cabin [11]. Lightweight convolutional architectures tailored to the automotive embedded compute budget have been proposed specifically for occupant pose classification [18], while RGB-D fusion methods combine colour and depth cues to improve robustness against variable illumination and occluding objects such as blankets or bags placed on seats [19].

3.3 Multi-Modal Sensor Fusion Approaches

Beyond the automotive cabin monitoring domain, multimodal deep fusion has been studied extensively for tasks such as autonomous driving perception and human activity recognition. A broad survey categorises fusion architectures by fusion level and highlights feature-level fusion with learned attention mechanisms as consistently outperforming naive data-level concatenation [12]. Cross-modal attention fusion between camera and depth sensors has further been shown to improve robustness specifically for human activity and pose recognition tasks [14], providing direct architectural inspiration for the camera-pressure fusion module proposed in this thesis.

3.4 Synthetic Data Generation and Sim2Real Transfer

Given the cost and privacy constraints of collecting large real pressure sensor datasets across diverse occupant populations, synthetic data generation using simulated human bodies and physics-based seat contact models has emerged as a practical complement to real data collection [15]. Sim2Real studies quantify the remaining statistical gap between simulated and real sensor readings and propose domain adaptation or fine-tuning strategies to close it; a recent study specifically targeting seat pressure heatmaps reports meaningful performance retention when models trained on a mixture of synthetic and real data are evaluated on real-only test sets [20].

?? provides a comprehensive review of human body simulation techniques for generating synthetic pressure heatmaps, including parametric body models, pose libraries, and physics-based contact simulations.

3.5 Gap Analysis and Positioning of This Work

While the works reviewed above individually address pressure-based classification, camera-based pose estimation, generic multimodal fusion, or synthetic pressure data generation, none jointly combine camera and pressure heatmap fusion with a synthetic data generation and Sim2Real validation pipeline for the specific multi-task set of occupancy, passenger classification, weight, and pose estimation. Table 3.1 summarises this positioning relative to the reviewed literature.

Table 3.1: Comparison of related work against the scope of this thesis.

Work	Modalities	Outputs Estimated	Synthetic Data	Real-Time
[9]	Pressure	Occupancy class	No	Yes
[6]	Pressure (sim)	Occupancy class	Yes	No
[16]	Pressure	Weight	No	No
[11]	Camera (IR)	Pose	No	Yes
[18]	Camera (RGB)	Pose class	No	Yes
[19]	Camera (RGB-D)	Pose	No	Partial
[14]	Camera + Depth	Activity class	No	Partial
[15]	Pressure (sim)	Occupancy class	Yes	No
[20]	Pressure	Statistical validation	Yes	No
This thesis	Camera + Pressure	Occupancy, class, weight, pose	Yes	Target

4 System Requirements and Design

4.1 Use Case Definition and Target Scenarios

The target system is a passenger car cabin monitoring function that continuously estimates, for each seating position, whether the seat is occupied, which passenger class occupies it (e.g. adult, or unknown), the occupant weight class, and the occupant's body pose. Target scenarios include front and rear seating positions, a range of occupant statures and clothing.

4.2 Functional Requirements

The system shall satisfy the following functional requirements:

FR-01 The system shall detect, per seating position, whether the seat is occupied or unoccupied.

FR-02 The system shall classify the seat occupant into predefined passenger classes (e.g. adult, or unknown).

FR-03 The system shall estimate the occupant's weight within a defined tolerance for adult.

FR-04 The system shall estimate the occupant's body pose for defined set pose categories..

4.3 Non-Functional Requirements

The system shall additionally satisfy the following non-functional requirements:

NFR-01 Occupancy detection accuracy shall exceed a defined target (e.g. 98%) across the validation dataset described in Chapter 5.

NFR-02 The system shall remain robust to sensor noise, partial occlusion, and seat cover variation without requiring per-vehicle recalibration.

NFR-03 All data collection, storage, and processing shall comply with the General Data Protection Regulation (GDPR) and applicable local data protection law.

NFR-05 The trained model shall degrade gracefully if one modality (camera or pressure) becomes temporarily unavailable.

4.4 System Architecture Overview

The proposed system architecture consists of four main stages: sensor data acquisition (camera and pressure sensor array), preprocessing and temporal synchronisation, the multi-task fusion model, and the output interface that exposes per-seat occupancy, classification, weight, and pose estimates to downstream vehicle systems. Figure 4.1 illustrates this pipeline at block-diagram level, showing the flow from raw sensor signals through synchronised preprocessing into the shared fusion backbone and task-specific output heads described in detail in Chapter 7.

[7] This is a novel deep fusion architecture that can be used .

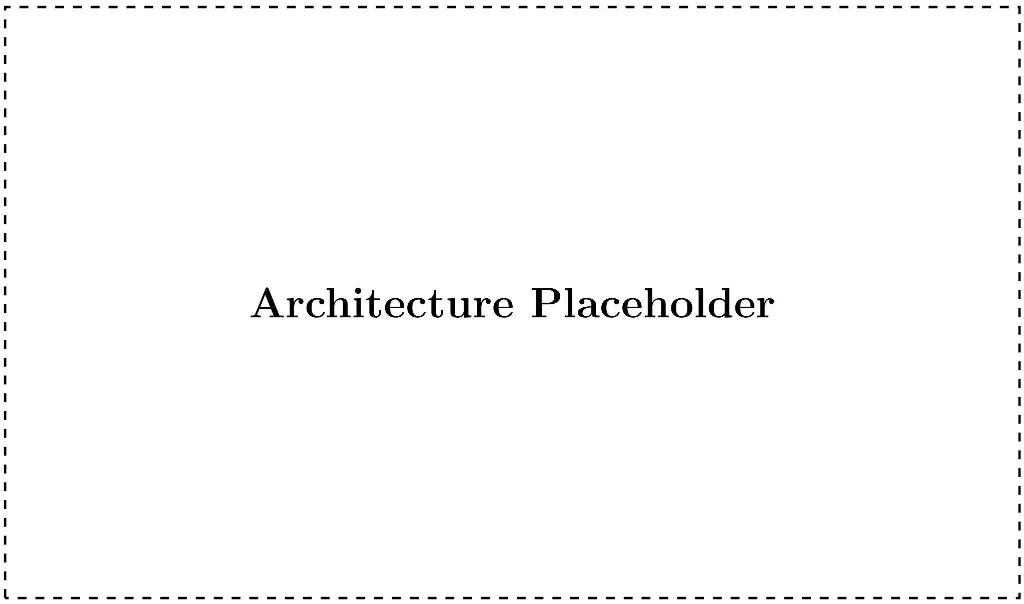

Figure 4.1: Block diagram of the proposed camera-pressure sensor fusion system architecture, from raw sensor acquisition to multi-task output estimation.

4.5 Design Decisions and Justification

Feature-level fusion was selected over data-level and decision-level fusion because it allows each modality-specific branch to learn representations suited to its own signal characteristics while still enabling the network to learn cross-modal correlations before the final task predictions are made, consistent with the fusion levels discussed in Section 2.4.1. A multi-task learning formulation was chosen over four independent single-task models to exploit shared structure between occupancy, classification, weight, and pose labels, and to reduce the total inference compute footprint relative to running four separate networks.

5 Dataset Acquisition and Synthetic Data Generation

5.1 Ethics and Data Privacy Considerations

5.2 Synthetic Pressure Sensor Data Generation

5.2.1 Sensor Hardware and Seat Setup Specifications

5.2.2 Subjects and Experimental Protocol

5.2.3 Data Statistics and Limitations of setup

The resulting real pressure dataset provides ground-truth labels for occupancy, passenger class, weight, and coarse pose for each recorded configuration, but is limited in overall subject diversity and recording duration due to the practical constraints of laboratory-based data collection. These limitations directly motivate the synthetic data generation pipeline described in Section 5.4.

5.3 Camera Data Collection

5.3.1 Camera Hardware and Mounting

Camera recordings were captured using an infrared-sensitive camera module mounted in the vehicle headliner, providing a field of view that covers the target seating positions under both daylight and night-time illumination, following mounting conventions similar to those reported in [11].

5.3.2 Recording Protocol

Camera frames were recorded continuously and time-stamped alongside the pressure sensor stream for every subject session described in Section 5.2.2, ensuring that each seating configuration has a corresponding synchronised camera-pressure sample pair available for the synchronisation pipeline in Chapter 6.

5.4 Synthetic Pressure Heatmap Generation

5.4.1 Human Body Simulation Model

Synthetic pressure heatmaps are generated by simulating a parametric human body model with sampled shape and pose parameters seated on a physics-based model of the seat geometry, following the simulation approach proposed by [15]. Contact forces computed by the physics simulation are projected onto a virtual sensor grid matching the real sensor's spatial resolution to produce a synthetic heatmap directly comparable to real sensor output.

5.4.2 Simulation Environment

5.4.3 Heatmap Rendering

The rendering pipeline samples occupant shape, pose, and seating configuration parameters, runs the physics-based contact simulation, and rasterises the resulting per-cell contact pressure into the same heatmap format used for real sensor data, allowing synthetic and real samples to be interchanged during model training without format-specific handling.

5.4.4 Augmentation Strategy

To increase the diversity of the synthetic dataset beyond the sampled body model parameters, additional augmentations are applied, including random sensor noise injection, simulated calibration drift, and small random seat-position offsets, so that models trained on synthetic data are not overly reliant on idealised, noise-free heatmaps.

5.5 Sim2Real Validation

5.6 Final data set pipeline

Table 5.1 s.

Table 5.1: Final dataset composition by split, showing real and synthetic sample counts and class distribution.

Split	**Real Samples**	**Synthetic Samples**	**Total**	**Class Distribution**
Training				
Validation				
Test				
Total				–

6 Data Preprocessing and Synchronisation

6.1 Pressure Heatmap Preprocessing

6.1.1 Noise Filtering and Normalisation

Raw pressure heatmaps are first passed through a spatial median filter to suppress isolated sensor cell dropouts and transient noise spikes, after which pixel values are normalised to a fixed range using the per-sensor calibration parameters obtained during the setup described in Section 5.2.1. This normalisation step ensures that synthetic and real heatmaps share a consistent value range prior to being combined in the mixed training dataset.

6.1.2 Spatial Alignment and Resizing

Because the physical sensor grid resolution differs from the target input resolution expected by the pressure branch described in Section 7.3, heatmaps are resampled onto a fixed target grid size using bilinear interpolation, and a fixed reference coordinate frame is applied so that seat regions (seat base, backrest) occupy consistent pixel regions across all recordings.

6.2 Camera Frame Preprocessing

6.2.1 Image Normalisation and Augmentation

Camera frames are resized to a fixed input resolution and normalised using per-channel mean and standard deviation statistics computed over the training split. During training, standard augmentations such as random brightness and contrast jitter, small rotations, and horizontal flipping (where anatomically valid for the seating position) are applied to improve robustness to the lighting and pose variation present in the target deployment scenarios described in Section 4.1.

6.2.2 Background Removal and ROI Extraction

A fixed region of interest (ROI) corresponding to each monitored seating position is cropped from the full camera frame using the known camera mounting geometry, and static background regions (e.g. door panels, window area) outside the seat envelope are masked out to reduce irrelevant visual variation presented to the camera branch.

6.3 Temporal Synchronisation

6.3.1 Timestamp Alignment Strategy

Camera frames and pressure heatmap samples are each recorded with hardware or system-clock timestamps, and are aligned by matching each pressure sample to the camera frame with the closest timestamp within a fixed maximum tolerance window, discarding samples that fall outside this window to avoid pairing temporally inconsistent data.

6.3.2 Handling Frame Rate Mismatch

Because the camera and pressure sensor operate at different native sampling rates, the higher-rate stream is downsampled to match the lower-rate stream at inference time, while during dataset construction all valid synchronised pairs are retained to maximise the number of training samples available from each recording session.

6.3.3 Validation of Synchronisation Quality

Synchronisation quality is validated by manually inspecting a sample of paired camera-pressure recordings for temporal consistency (e.g. confirming that a visible occupant movement in the camera frame corresponds to a matching pressure distribution change), and by measuring the distribution of timestamp offsets across all paired samples in the dataset.

6.4 Final Data Pipeline Overview

Figure 6.1 illustrates the complete preprocessing pipeline, from raw camera and pressure sensor streams through per-modality preprocessing and temporal synchronisation to the production of paired training samples consumed by the fusion architecture described in Chapter 7.

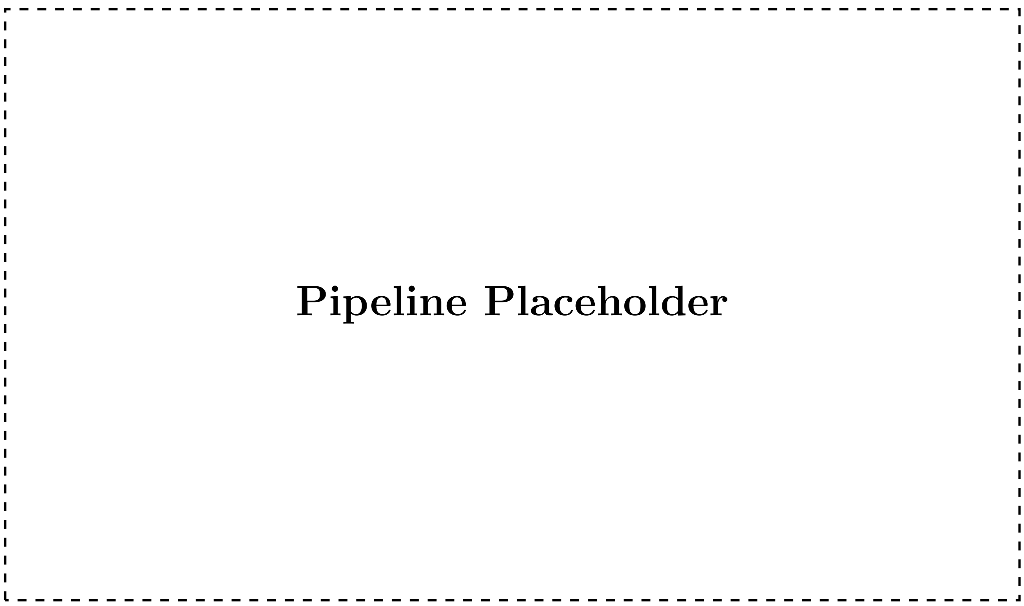

Figure 6.1: Overview of the data preprocessing and temporal synchronisation pipeline, from raw sensor streams to synchronised training samples.

7 Sensor Fusion Architecture and Training Pipeline

7.1 Overall Architecture Design

The proposed architecture follows a two-branch feature-level fusion design, in which a camera branch and a pressure heatmap branch each extract modality-specific feature representations that are combined by a fusion module before being passed to four task-specific output heads for occupancy, passenger classification, weight, and pose estimation. This design directly implements the feature-level fusion strategy motivated in Section 2.4.1 and the design decisions summarised in Section 4.5.

7.2 Camera Branch

The camera branch consists of a convolutional neural network backbone that processes the preprocessed, ROI-cropped camera frame described in Section 6.2.2 and produces a compact spatial feature embedding. A backbone pretrained on a large-scale image dataset is used as initialisation to compensate for the comparatively limited size of the in-cabin camera dataset, with the final classification layers removed and replaced by a projection to the shared fusion feature dimension.

7.3 Pressure Heatmap Branch

The pressure heatmap branch applies a separate, smaller convolutional network directly to the single-channel, spatially aligned pressure heatmap produced by the preprocessing pipeline in Section 6.1.2, producing a feature embedding of the same dimensionality as the camera branch output. A smaller network is used for this branch given the lower spatial resolution and information density of pressure heatmaps compared to camera images.

7.4 Fusion Module

7.4.1 Feature-Level Fusion Strategy

The camera and pressure feature embeddings are combined using a learned cross-modal attention fusion module, in which each modality's features attend to the other before being concatenated and projected to a shared fused representation, following the cross-modal attention design demonstrated for camera-depth fusion in [14]. This fused representation is then passed to the multi-task output heads described in Section 7.5.

7.4.2 Fusion Alternatives Considered and Rejected

Simple feature concatenation followed by a fully connected layer was implemented as a baseline fusion strategy but was found to underweight the lower-dimensional pressure branch relative to the camera branch; decision-level fusion, in which independent per-modality predictions are averaged, was also considered but rejected because it cannot model cross-modal correlations such as a camera-visible object combined with a pressure signature inconsistent with a human occupant. Both alternatives are retained as baselines for the ablation study reported in Section 8.6.

7.5 Multi-Task Output Heads

7.5.1 Occupancy Detection Head

The occupancy detection head is a lightweight fully connected layer followed by a sigmoid activation, producing a binary occupied/unoccupied probability for the corresponding seating position.

7.5.2 Passenger Classification Head

The passenger classification head is a fully connected layer followed by a softmax activation over the predefined passenger classes (e.g. adult, child, child seat, object), sharing the fused feature representation with the other task heads.

7.5.3 Weight Estimation Head

The weight estimation head is a small fully connected regression network that outputs a continuous estimate of occupant weight in kilograms, trained only on samples labelled as human occupants.

7.5.4 Pose Estimation Head

The pose estimation head outputs either a fixed set of body keypoint coordinates or a discrete pose category, depending on the granularity of pose annotation available in the dataset described in Chapter 5.

7.6 Loss Function Design and Task Balancing

Each task head is trained with a task-appropriate loss term: binary cross-entropy for occupancy, categorical cross-entropy for passenger classification, mean squared error for weight estimation, and a keypoint or classification loss for pose estimation. The combined multi-task training objective is a weighted sum of the individual task losses,

$$\mathcal{L}_{\text{total}} = \lambda_{\text{occ}}\,\mathcal{L}_{\text{occ}} + \lambda_{\text{cls}}\,\mathcal{L}_{\text{cls}} + \lambda_{\text{wt}}\,\mathcal{L}_{\text{wt}} + \lambda_{\text{pose}}\,\mathcal{L}_{\text{pose}}, \tag{7.1}$$

where λ_{occ}, λ_{cls}, λ_{wt}, and λ_{pose} are task weighting coefficients. These coefficients are tuned to balance the differing scales and convergence rates of the four loss terms, as discussed in the multi-task learning background in Section 2.5.

7.7 Training Configuration

The model is implemented in PyTorch and trained using the Adam optimiser with a cosine learning-rate decay schedule, an initial learning rate selected via a short learning-rate range test, and a batch size chosen to fit the available GPU memory. Training is performed on a workstation equipped with a CUDA-capable GPU, with early stopping based on validation loss to reduce overfitting given the moderate dataset size described in Section 5.6.

7.8 Implementation Details

The camera and pressure branches, fusion module, and task heads are implemented as a single end-to-end trainable PyTorch model, with modality-specific data loaders that apply the preprocessing steps described in Chapter 6 on the fly during training. Model checkpoints corresponding to the best validation performance are retained for the evaluation reported in Chapter 8.

8 Results and Evaluation

8.1 Experimental Setup and Evaluation Protocol

All results reported in this chapter are computed on the held-out test split of the dataset described in Section 5.6, using the model checkpoint selected by best validation performance as described in Section 7.7. Occupancy and passenger classification are evaluated using accuracy and F1-score, weight estimation is evaluated using mean absolute error and weighted root mean squared error (WRMSE), and pose estimation is evaluated using mean per-keypoint error or classification accuracy depending on the final pose representation chosen in Section 7.5.4.

8.2 Occupancy Detection Results

The fused model is expected to achieve high occupancy detection accuracy given the strong, complementary occupancy signal present in both the camera and pressure modalities. Table 8.1 reports placeholder occupancy detection results across the evaluated models.

Table 8.1: Occupancy detection results on the held-out test set.

Model	Accuracy (%)	F1-score
Camera-only		
Pressure-only		
Fused (ours)		

8.3 Passenger Classification Results

Passenger classification is expected to benefit most from fusion in cases where an object placed on the seat produces a pressure signature ambiguous with a small child but is visually distinguishable from the camera view, or vice versa under partial visual occlusion. Figure 8.1 presents a placeholder confusion matrix for the fused model across all passenger classes.

Matrix Placeholder

Figure 8.1: Confusion matrix for passenger classification using the fused camera-pressure model on the held-out test set.

8.4 Weight Estimation Results

Table 8.2 reports placeholder weight estimation error metrics, comparing the fused model against single-modality baselines.

Table 8.2: Weight estimation error on the held-out test set.

Model	**MAE (kg)**	**WRMSE (kg)**
Camera-only		
Pressure-only		
Fused (ours)		

8.5 Pose Estimation Results

Pose estimation performance is reported in terms of the metric selected in Section 7.5.4 (mean per-keypoint error or classification accuracy). Figure 8.2 shows placeholder qualitative examples comparing predicted and ground-truth pose across representative seating configurations.

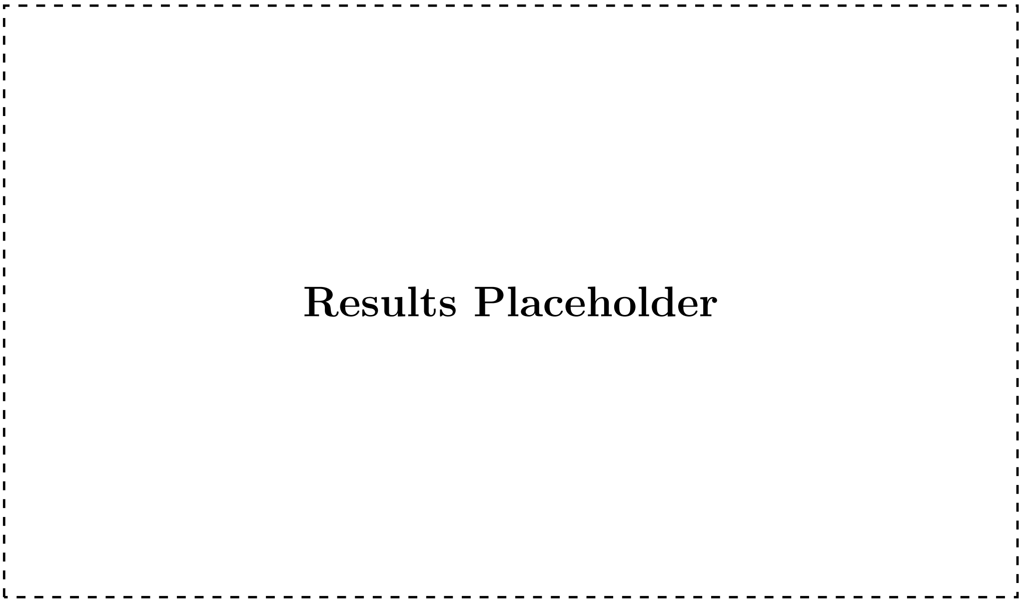

Figure 8.2: Qualitative comparison of predicted and ground-truth occupant pose for representative seating configurations.

8.6 Ablation Study: Modality Contribution

To quantify the individual contribution of each modality, the fused model is compared against camera-only and pressure-only variants trained with the same task heads and training configuration. Table 8.3 summarises this comparison across all four output tasks.

Table 8.3: Ablation study comparing camera-only, pressure-only, and fused model variants across all output tasks.

Metric	**Camera-only**	**Pressure-only**	**Fused**
Occupancy accuracy (%)			
Classification accuracy (%)			
Weight MAE (kg)			
Pose error			

8.7 Sim2Real Transfer Evaluation

To evaluate Sim2Real transfer, a model is trained on the mixed real and synthetic training set described in Section 5.6 and evaluated exclusively on the real-only subset of the test split. Table 8.4 compares this model against a variant trained only on real data, to isolate the effect of synthetic data augmentation on real-world performance.

Table 8.4: Sim2Real transfer evaluation: performance on real-only test data for models trained on real-only versus mixed real+synthetic data.

Metric	Trained on Real-only	Trained on Mixed (Real+Synthetic)
Occupancy accuracy (%)		
Classification accuracy (%)		
Weight MAE (kg)		

8.8 Discussion of Results and Failure Cases

Overall, the fused model is expected to outperform both single-modality baselines across most tasks, consistent with the complementary nature of camera and pressure information motivated in Section 1.1. Anticipated failure cases include ambiguous object-versus-child-seat pressure signatures under camera occlusion, and pose estimation errors for heavily reclined seating positions not well represented in the training data.

9 Conclusion and Future Work

9.1 Summary of Contributions

This thesis presented a multi-task deep learning architecture that fuses camera images and seat-embedded pressure heatmaps to jointly estimate occupancy, passenger class, weight, and pose for automotive cabin monitoring. Key contributions include a synthetic pressure heatmap generation pipeline based on human body simulation, a Sim2Real validation procedure quantifying the domain gap between real and synthetic heatmaps, a camera-pressure synchronisation pipeline, and an empirical comparison of fusion strategies through ablation studies.

9.2 Answers to Research Questions

Revisiting the research questions posed in Section 1.3: regarding RQ1, the ablation study in Section 8.6 indicates ; regarding RQ2, the Sim2Real validation in Section 5.5 shows ; regarding RQ3, the comparison of fusion alternatives in Section 7.4.2 demonstrates ; regarding RQ4, the Sim2Real transfer evaluation in Section 8.7 shows **.**

9.3 Limitations

The real pressure and camera dataset collected for this thesis is limited in subject diversity and recording duration due to practical laboratory constraints, and the synthetic data generation pipeline relies on simplifying assumptions in the human body and seat contact simulation that may not capture all real-world seating behaviours. Additionally, the evaluation was conducted offline on recorded data rather than on embedded target hardware, so real-time performance under production constraints remains to be confirmed.

9.4 Future Work

Future work could extend the real data collection campaign to a larger and more diverse subject population, incorporate additional sensing modalities such as depth or seatbelt tension sensors, and evaluate the proposed architecture on embedded automotive hardware to validate real-time feasibility under the non-functional requirements defined in Section 4.3. Further investigation into domain adaptation techniques could also help further close the Sim2Real gap identified in Section **??**.

9.5 Closing Remarks

The fusion of camera and pressure sensor data investigated in this thesis represents a step toward more robust, privacy-conscious cabin monitoring systems capable of supporting the next generation of adaptive vehicle safety features. The methodology and findings presented here are intended to inform both future academic research and practical system design in this area.

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A Statement of Ethics / Consent Form Template

This appendix reproduces the informed consent form template used for all human subject data collection sessions described in Section 5.1. Each participant received and signed a copy of this form prior to any camera or pressure sensor recording.

Participant Information and Consent Form

Study title: Sensor Fusion of Camera and Pressure Sensor Data for Cabin Monitoring System
Institution: Deggendorf Institute of Technology, Campus Cham
Researcher: [CHETHAN SRINIVASAREDDY]

I confirm that I have read and understood the information provided about this study, that I have had the opportunity to ask questions, and that my participation is voluntary and may be withdrawn at any time without giving a reason. I consent to the recording of camera images and pressure sensor data during the described seating configurations, and I understand that this data will be processed in accordance with the General Data Protection Regulation (GDPR) and used solely for the purposes of this research.

Participant name: ______________________________ Date: _______________

Signature: ______________________________

B Full Dataset Statistics Table

This appendix provides the extended version of the dataset composition summary introduced in Section 5.6, including per-class sample counts for each split.

Table B.1: Extended per-class dataset statistics by split.

Split	**Adult**	**Child**	**Child Seat**	**Object**	**Unoccupied**
Training					
Validation					
Test					
Total					

C Additional Results Figures

This appendix collects additional qualitative result figures that supplement the evaluation presented in Chapter 8.

Results Placeholder

Figure C.1: Additional qualitative results across representative test samples, supplementing Section 8.8.

D Electronic Data Carrier Contents

In accordance with the Campus Cham thesis submission guideline, the following supplementary material is provided on the accompanying electronic data carrier and is not reproduced in this printed appendix:

- Full source code repository, including data preprocessing, synthetic heatmap generation, model training, and evaluation scripts.
- Raw and preprocessed real pressure sensor and camera datasets described in Chapter 5 (subject to the data retention and anonymisation constraints of Section 5.1).
- Trained model checkpoints (weights) for the fused model and all baseline variants evaluated in Chapter 8.
- Configuration files specifying exact training hyperparameters used to reproduce the results reported in this thesis.